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Education

University of Michigan - Ann Arbor

Expected May 2027

Bachelor of Science in Computer Science & Biopsychology, Cognition, and Neuroscience (GPA: 3.89/4.00)

Ann Arbor, MI

- **Honors:** Rogel Scholar (Full-Ride Scholarship), William J. Branstrom Prize (Top 5%), James B. Angell Scholar (5 Straight-A Terms)
- **Relevant Coursework:** Data Structures & Algorithms, Discrete Math, Computer Organization, Foundations of Computer Science, Machine Learning, Web Systems, Applied Linear Algebra, Statistics & Data Analysis

Experience

Sartorius

May 2026 – Aug 2026

SDE Intern, Advanced Algorithms Team

Ann Arbor, MI

- Engineered 2 C++ pipelines to deploy pretrained ML models into production, connecting model inference to image fluorescence quantification and whole-image masking calculations through to the application layer.
- Eliminated circular dependencies across internal team libraries by architecting a shared interface system of flattened DTOs for 17 analyses, improving build reliability and enabling independent cross-team development.
- Redesigned a monolithic Qt image analysis tool into a modular MVC system with per-module controllers, decoupling UI (Qt Designer) from logic (Python) across 5 modules and adding multithreading, cross-window state sync, and error handling.
- Consolidated 7 fragmented Conda environments into a single modular environment, implementing conda-lock for reproducible builds and custom tooling for environment setup, sync, and validation in Python.

Projects

American Airlines Sponsored Project Team

Jan 2026 – Apr 2026

Student Machine Learning Engineer, Michigan Data Science Team

- Developed a hybrid ML pipeline with Random Forest, XGBoost, and GenAI to minimize high-cost flight simulator cancellations from historical operational logs, achieving 85.2% accuracy in predicting sequence outcomes.

Replica of GPT-2

Sep 2025 – Dec 2025

Michigan Data Science Team

- Implemented a GPT-2-style Transformer language model from first principles in PyTorch, building self-attention, multi-head causal attention, MLP blocks, and layer normalization without high-level abstractions.
- Trained model on real-world text corpora (Shakespeare, Wikipedia) to generate coherent text, and validated architectural correctness by benchmarking against pretrained GPT-2 weights.

AI-Powered Student Dashboard

June 2026 - Aug 2026

Personal

- Built a full-stack web application with FastAPI and React, integrating an NLP-based chatbot to answer student queries and generate personalized workflow recommendations in real time.
- Designed SQLite-backed REST APIs and optimized frontend state management for low-latency interaction and data visualization.

Research

Computational & Cognitive Neuroscience Lab, University of Michigan

Aug 2024 – Present

Research Assistant

Ann Arbor, MI

- Build R pipelines for statistical analysis and visualization of multimodal neural and behavioral data across hundreds of samples.
- Perform quality control on MRI/MRS data and manual FreeSurfer segmentation edits on structural brain images.
- Conduct behavioral data entry and preprocessing to support longitudinal cognitive decline analyses.
- Poster (UM Undergraduate Research Symposium, 2024): Investigating Age-Related Neural Dedifferentiation Longitudinally and in Mild Cognitive Impairment.
- Honors Thesis (submission: April 2027): "Predicting Cognitive Decline from Multimodal Neural and Behavioral Data: A Machine Learning Approach to Age-Related Neural Dedifferentiation".

Regenerative Imaging Laboratory, University of Pittsburgh

May 2025 – Aug 2025

Research Fellow

Pittsburgh, PA

- Analyzed hippocampal connectivity in 18 brains using diffusion MRI and tractography, performing segmentation in DSI Studio.
- Applied k-means clustering in MATLAB to model neural connection patterns, identifying structural network trends.
- Final Presentation (Center for Neuroscience, 2025): "Segmentation of hippocampal dentations and virtual dissection of their connectome using mesoscale diffusion MRI".

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, C, R

Frameworks/Libraries: React, FastAPI, PyTorch, scikit-learn, Qt, OpenCV, Svelte

Tools: Docker, Git, Conda, MATLAB, DSI Studio, Streamlit

ML/DS: Transformers, CNNs, ResNet, Random Forest, XGBoost, k-means Clustering, Transfer Learning